

MODULE 2: WATER & FOOD

2.1. OBTAINING AND TREATING WATER

Water from natural sources may contain biological contaminants such as bacteria, viruses, and protozoa, as well as chemical contamination. Drinking untreated water can cause severe gastrointestinal disease and accelerate dehydration.

Biological contaminants

- **Protozoa:** Giardia lamblia and Cryptosporidium; resistant cysts.
- **Bacteria:** Escherichia coli, Salmonella, Vibrio cholerae.
- **Viruses:** rotaviruses and hepatitis A; small enough to pass through many mechanical filters.

Water-treatment methods

Method	Effectiveness	Procedure / requirements
Boiling	Biological contaminants	Bring water to a full rolling boil. The source specifies at least 1 minute at sea level or 3 minutes above 2,000 m.
Chemical disinfection	High; weaker against Cryptosporidium	The source discusses chlorine-dioxide products or iodine and notes that reaction time increases in cold water. Does not remove chemical toxins.
Mechanical filtration	High for bacteria and protozoa	Filters with pores around 0.1 micron are described; additional treatment may be needed for viruses.
Improvised gravity filter	Pre-treatment / clarification only	Layers of gravel, fine sand, charcoal, sand and cloth can reduce turbidity and odour but do not reliably remove microbes.

2.2. FORAGING

Gathering plants can require less energy than hunting but carries a serious poisoning risk. The manual's core rule is: never eat a plant unless you are completely certain of its identification.

The manual advises avoiding unfamiliar plants with features such as milky or coloured sap, strongly bitter or burning taste, white/yellow berries, almond/peach-like odours, and members of the carrot/parsley family unless identification is certain.

Universal Test of Edibility

The source presents a multi-stage test intended only for extreme situations when no alternative food is available. It involves fasting, skin contact, lip and tongue contact, chewing, swallowing a very small amount, and long observation periods. Because some toxins are not reliably detected by such tests, positive plant identification remains safer than relying on a generic edibility test.

2.3. FISHING & TRAPPING

The manual notes that poaching and trapping are illegal in normal peacetime conditions, while discussing them in the context of extreme survival.

- **Trigger mechanisms:** Figure-4 and spring-pole style triggers are described.
- **Snares:** may be made from wire or cord and placed on actively used animal trails.
- **Location:** fresh game trails, burrow entrances, and feeding areas are identified as likely locations.

2.4. SAFE FOOD STORAGE

Raw or cooked food odours can attract wildlife. In bear country the manual describes a bear-bag approach:

- Pack food, cosmetics, and dirty cookware in a waterproof bag.

- Hang the bag at least about 4 m above the ground and about 2 m away from the tree trunk.
- Keep food storage and cooking areas roughly 100 m away from the sleeping area, forming a camp triangle.